Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653146809
Question Shuffling Allowed : No

Question Number : 100 Question Id : 640653995531 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :
6406533360882. ✔ YES
6406533360883. ✖ NO

Sub-Section Number : 2
Sub-Section Id : 640653146810
Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 640653995532 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

Options :
6406533360884. He should buy the earphones which have high latency as much possible.
6406533360885. It all depends on Internet Speed, so we can’t determine anything.
6406533360886. He should buy the earphones which have low latency as much possible.
6406533360887. He should buy the earphones which have as low Bluetooth version as possible.

Question Number : 102 Question Id : 640653995543 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Consider the below URL:

https://www.google.com/search?q=physics+books

protocol Host query string

Select the correct answer that identifies the different components of the given URL.

Options :

Host: http
Protocol: google.com/search
Query string: q=physics+books

Protocol: httpS
Host: google.com/search
Query string: q=physics+books

request.args.get("q")

Protocol: http
Query string: google.com/search
Protocol: q=physics+books

Host: http
Query string: google.com/search
Protocol: q=physics+books

Sub-Section Number : 3
Sub-Section Id : 640653146811
Question Shuffling Allowed : Yes

Question Number : 103 Question Id : 640653995533 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following HTML code and select the correct rendered output from the following.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
      <tr>
        <th>One</th>
        <td>Four</td>
      </tr>
      <tr>
        <th> Two</th>
        <td> Three</td>
      </tr>
    </table>
</body>
</html>
```

One | Four

Table-Head

row

Head-Two | Three

Options :

✓

One	Four
Two	Three

One	**Two**
Four	Three

One	**Two**
Three	Four

One	**Four**
Two	Three

Question Number : 104 Question Id : 640653995534 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following code block and select the correct option regarding the output when we run the python file app.py.

app.py

jinja2 → {{ }}

X jinja syntax

String → $

```
from jinja2 import Template

template='''
          Welcome to {Degree} DEGREE
          First level to clear is {{level}}
          '''

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

Options :

Welcome to {Degree} DEGREE
First level to clear is Foundation

Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

Welcome to {Degree} DEGREE
First level to clear is

Welcome to IITM BS DEGREE
First level to clear is Foundation

Welcome to DEGREE
First level to clear is {{ Level }}

Question Number : 105 Question Id : 640653995538 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The following Python code snippet generates the output on the terminal.

```
from string import Template
greet_template = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)
```

Hello, Alico! Welcome to Wonderland

```
data = dict(name='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)
```

Hello, Bob! Welcome to $place.

```
greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of the following is the correct output?

Options :

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Question Number : 106 Question Id : 640653995540 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

The directory listing of a folder and the content of each of its files are given below.

Directory Listing:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

Filename: base.html

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

Filename: home.html

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

Filename: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

python -m http.server

Files

If a simple Python HTTP server is created for this directory, what will be rendered on the browser for base URL?

if index file is not in directory

Options :

Welcome to PDSA course

Welcome to DBMS course → index.html

Welcome to MAD-I course

Directory listing for /

Question Number : 107 Question Id : 640653995544 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

    def __init__(self, accno, loan_amount, interest_rate):
        self.accno = accno
        self.loan_amount = loan_amount
        self.interest_rate = interest_rate

    def get_loan_details(self):
        return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

loan1.get_loan_details()

Options :

Account1 : (102, 170000, 6)
Account2 : (101, 45900, 5)

Account2 : (102, 170000, 6)

Account1 : (101, 45900, 5)

Account1 : (101, 45900, 5)
Account2 : (102, 170000, 6)

Question Number : 108 Question Id : 640653995548 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following flask view functions will return a 404 error for the URL:

http://127.0.0.1:5000/communication/landline/0712 ?

Online ?

Options :

landline/0712

```
@app.route('/communication/<type>/<areacode>')
def home(type, areacode):
    details = {'type': type, 'area-code': areacode}
    return details
```

×404

str str

Best option

6406533360946.

```
@app.route('/communication/<string:type>/<int:areacode>')
def home(type, areacode):
        details = {'type': type, 'area-code': areacode}
    return details
```

×404

int(0712) → 712

6406533360947.

```
@app.route('/communication/<string:type>/0712')
def home(type):
    details = {'type': type, 'area-code': 044}
    return details
```

×404

6406533360948.

```
@app.route('/communication/online/<string:type>/<areacode>')
def home(type, areacode):
    details = {'type': type, 'area-code': 044}
    return details
```

Sub-Section Number : 4
Sub-Section Id : 640653146812
Question Shuffling Allowed : Yes

Question Number : 109 Question Id : 640653995539 Question Type : MCQ
Correct Marks : 4.5

Mega

bits/second

Question Label : Multiple Choice Question

A certain network connection with a bandwidth of $\boxed{10 \text{ Mbps}}$ allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

Options :

6406533360908. 18 Mbps
6406533360909. 16 Mbps
6406533360910. 26 Mbps
6406533360911. 36 Mbps

Question Number : 110 Question Id : 640653995541 Question Type : MCQ
Correct Marks : 4.5

Question Label : Multiple Choice Question

Mega 10^6

Kilo 10^9

Bandwidth =

$$\text{Maximum Size} = \frac{\text{Bandwidth}}{\text{requests}} = \frac{10 \times 10^6}{\underset{10^3}{1000}} = 10 \times 10^3 = 10\ \text{kb}$$

$$10\text{kb} + 8\text{kb} = \text{data size} = 18\ \text{kb}$$

$$18\ \text{kb} = \frac{\text{bandwidth}}{2000}$$

$$\text{bandwidth} = 18 \times 10^3 \text{ bits} \times \underbrace{10^3 \times 2}_{2000}$$

$$= 36 \times \underbrace{10^6}_{\text{Mega}} \text{ bit}$$

$$= 36\ \text{Mbps}$$

New

Consider the following Python code snippet.

Filename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

print(sys.argv[1:])

argv

If the above flask app is run using the command `python app.py GET POST` in one terminal, what will be the output on another terminal for command:

0 1 2

`curl http://127.0.0.1:5000/home` ? GET

Options :

<h1>Hello from GET</h1>

<h1>Hello from POST</h1>

<h1>Please enter a valid HTTP method</h1>

<h1>Method not Allowed</h1>

Question Number : 111 Question Id : 640653995546 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would ________.

Copper

[The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

$C \leftarrow S$

Options :

6406533360937. Increase

6406533360938. Decrease

6406533360939. Remain Same

6406533360940. Insufficient Information

Sub-Section Number :	5
Sub-Section Id :	640653146813
Question Shuffling Allowed :	Yes

Question Number : 112 Question Id : 640653995542 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the below server response

version

HTTP/2.0 500 Internal Server Error

protocol error

Which of the following is/are **True**?

Options :

6406533360920. HTTP/2.0 is the protocol and version respectively

6406533360921. 500 is the port number

6406533360922. “Internal Server Error” is the status message

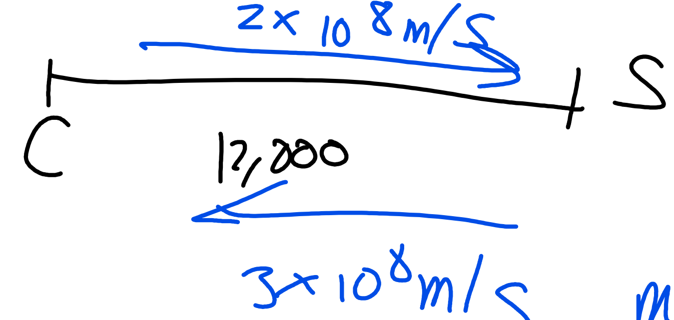
6406533360923. 500 is the status code

6406533360924. HTTP/2.0 is the server failure

Question Number : 113 Question Id : 640653995545 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question



S to C

millisec $\to 10^{-3}$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{12{,}000\ \text{Km}}{}$$

10^3

① Cable not broken

$$= \frac{12\times 10^6\ \text{m}}{2\times 10^8} = 6\times 10^{-2}\ \text{sec}$$

cable

$$= 60\ \text{msec}$$

$$= 120\ \text{msec}$$

② Cable broken, light

$$\frac{12\times 10^6\ \text{m}}{3\times 10^8} = 4\times 10^{-2}\ \text{sec}$$

light

$$= 40\ \text{msec} + 20\text{sec}$$

$$= 60\ \text{msec}$$

Consider the flask code below.

```
from flask import Flask

app = Flask(__name__)

@app.route("/employee/<id>")
def get_employee(id):
    return "Employee id : " + str(id)

if __name__ == "__main__":
    app.run(debug=True)
```

Assume that the above flask application is running on http://127.0.0.1:5000 . Which of the following URLs will render **"Not Found"** error?

Options :

http://127.0.0.1:5000/employee/102 Employee id : 102

http://127.0.0.1:5000/employee?id=102 request.args.get("id"): id=102 str(id)

http://127.0.0.1:5000/id=102 404

http://127.0.0.1:5000/102

Sub-Section Number : 6

Sub-Section Id : 640653146814

Question Shuffling Allowed : Yes

Question Number : 114 Question Id : 640653995547 Question Type : MSQ

Correct Marks : 4.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following code snippet.

```
@app.route('/name/<type>')
def prod_detail(type):
    # CODE BLOCK HERE
```

Assume the database has a "product" table represented by class "Product" which has a TEXT column "type". If we want the server to return a 404 status code when a user goes to the route '/name/<type>' with a "type" that does not exist in the database, which of the following lines would give us the desired output?

Options :

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html")
return render_template('details.html', type=type)
```

None

```
type = Product.query.filter_by(type=type).first()
if type is None:
    abort(404)
return render_template('details.html', type=type)
```

404 error in Terminal

all_or_404

```
type = Product.query.filter_by(type=type).first_or_404()
return render_template('details.html', type=type)
```

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html"), 404
return render_template('details.html', type=type)
```

Sub-Section Number : 7

Sub-Section Id : 640653146815

Question Shuffling Allowed : No

Question Id : 640653995535 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (115 to 116)

Question Label : Comprehension

Consider the following code and answer the given subquestions.

form.html

GET Apliation

```
<!DOCTYPE html>
<body>
      <form method=Method>
            Enter Your Name : <input type="text" name="name">
            <button>Submit</button>
      </form>
</body>
</html>
```

app.py

```
from flask import Flask
from flask import render_template, request

app=Flask(__name__)

@app.route("/",methods=['GET','POST'])

def home():
     if request.method=="GET":
            return render_template("form.html")
     else:
           return render_template("home.html")

app.run(debug=True)
```

Sub questions

Question Number : 115 Question Id : 640653995536 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Suppose the value of **Method** in form.html is GET (form method is set to GET) and in the input box we write “Application”, after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

http://127.0.0.1:5000/?name=Application

http://127.0.0.1:5000/name=Application

http://127.0.0.1:5000/?name

http://127.0.0.1:5000/

Question Number : 116 Question Id : 640653995537 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

http://127.0.0.1:5000/?name:Application

http://127.0.0.1:5000/?Application

http://127.0.0.1:5000/?name

http://127.0.0.1:5000/

Java

Section Id :	64065369407
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0